

Decryption Failures in NGCC Lattice KEMs: Correlated Blocks, Omitted Compression Noise, and Failure Boosting under a Query Cap

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September 2026

Abstract

The Chinese NGCC post-quantum competition received a family of lattice KEMs whose decryption failure rates (DFR) are certified by their designers with models that differ in detail. We recompute the failure probability of the first-round lattice KEMs whose claimed DFR, or a simple recomputation of it, lies near or below the nominal level, and we price the offline search for weak ciphertexts under a cap of 2^{64} or 2^{80} decapsulation queries. Three distinct mechanisms turn out to be responsible. (i) *Correlated decoding blocks*. DTRU decodes 16-coefficient DoubleE₈ blocks; in the tricyclotomic ring $\mathbb{Z}_q[x]/(x^n - x^{n/2} + 1)$ and the LPPNF ring the two octets of a block are ring-paired and the repetition code places the same bit on both. The exact maximum-likelihood failure region with the conditional block covariance raises the DFR of six of the seven sets by 50–110 bits; one failure costs 2^{85} – 2^{111} decapsulations against nominal levels of 128–512, and for DTRU-648, -768 and -Prime it fits into 2^{64} queries after 2^{22} – 2^{26} offline encapsulations. (ii) *Omitted compression noise*. Cheetah compresses its public key and both ciphertext components; Section 2.2 of the specification drops the public-key term $\langle e_b, r \rangle$ and subtracts the worst case of the v -compression error from the threshold. Evaluated by exponentially tilted one-dimensional FFTs, and validated on a scaled-down instance of Algorithms 15–17 and on full-size noise, the DFR of Cheetah128 and Cheetah256 is $2^{-79.1}$ and $2^{-51.8}$ instead of the claimed 2^{-129} and 2^{-176} : a random ciphertext fails within 2^{80} queries at both levels. (iii) *Bounded noise in two-dimensional codes*. Rudraksh2 and Scabbard decode pairs of coefficients with B2-Minal codes. The pair $(\Delta m_0, \Delta m_{n/2})$ is a sum of independent two-dimensional atoms; we evaluate its tail by one-dimensional projections onto the Voronoi radii of the real decoder. The exact DFR is $2^{-103.9}$ for Rudraksh2-128-I (claimed 2^{-100}) and $2^{-102.1}$ for Scabbard-128 (claimed 2^{-99}). Under a 2^{80} query cap Rudraksh2-128-I reaches a first failure after 2^{32} offline encapsulations, for a classical total of 2^{112} (below the level 128; the quantum total 2^{96} exceeds the 80-bit quantum target). Scabbard-128 without a cap costs 2^{102} , below 128, which is the comparison the specification itself makes against $2^{64} \times 2^{-99}$; with the cap the offline tilt costs 2^{108} and the classical total is 2^{188} . No attack we describe recovers a key; the results invalidate stated security levels in the DFR metric and the methods used to compute them.

1 Introduction

Lattice-based KEMs built from (Ring/Module-)LWE or NTRU decrypt correctly only with overwhelming probability. A small decryption failure rate (DFR) δ is harmless for correctness, but it enters the security of the Fujisaki–Okamoto transform through terms of the form $(q_H + q_D)\delta$, and, more concretely, an adversary who can trigger failures learns information about the secret

key. D’Anvers, Vercauteren and Verbaauwhede [6] introduced *failure boosting*: honestly generate many ciphertexts offline, keep those whose failure probability is unusually high, and submit only those to the decapsulation oracle. D’Anvers, Rossi and Virdia [7] showed that once a first failure is found, subsequent failures are much cheaper (*directional failure boosting*), so the security target is that even one failure must cost 2^λ . D’Anvers and Batsleer [8] refined the cost accounting and applied it to Saber and Kyber.

Designers therefore need a reliable DFR estimate. The common practice is to compute the distribution of a single error coefficient exactly (by convolution) and then to treat the coefficients that enter one decoding decision as independent. For schemes over $\mathbb{Z}_q[x]/(x^n + 1)$ with one bit per coefficient this is well studied [6, 8, 9]. Error-correcting codes change the picture: a decoding decision now depends on a block of coefficients, and any correlation inside the block changes the tail. Compression changes it too: the rounding error of every compressed object reappears in the decryption noise, and its bounded support makes Gaussian models inaccurate at the operating point.

Scope. We went through the lattice KEMs of the first NGCC round and selected those whose claimed DFR, or a simple recomputation of it, is below the nominal security level. For each we ask two questions: what is the failure probability of an honest ciphertext under the real decoder and the real noise, and how much offline work does an adversary need so that a first failure appears within 2^{64} or 2^{80} decapsulation queries. Two groups emerge. Class I comprises those sets for which $1/\mathbb{E}[\delta]$, or the classical total under a 2^{80} query cap, lies below the nominal level: the six DTRU sets over the tricyclotomic and LPPNF rings, and Cheetah128 and Cheetah256. Class II sits at the line: DTRU-Light, Rudraksh2-128-I (classical 2^{112} under the cap, quantum 2^{96} above the 80-bit quantum target) and Scabbard-128 (uncapped 2^{102} below 128; capped total 2^{188}). Schemes whose weakest legal ciphertext cannot reach a conditional failure probability of 2^{-80} , or whose offline search cost exceeds the level by hundreds of bits (FLIT, NTRE, BW-KEM, ZEN, NEV, the higher Cheetah, Rudraksh2 and Scabbard sets), are outside this paper.

DTRU. DTRU [1] is an NTRU-based KEM. Its main parameter sets live in the tricyclotomic ring $\mathcal{R}_q = \mathbb{Z}_q[x]/(x^n - x^{n/2} + 1)$ with $q = 3457$, a variant DTRU-Prime uses the LPPNF ring $\mathbb{Z}_q[x]/(x^{1087} - x - 1)$, and a lightweight set DTRU-Light uses $\mathbb{Z}_{769}[x]/(x^{512} + 1)$. Messages are encoded with DoubleEg: four bits become an $[8, 4, 4]$ extended Hamming codeword, which is then repeated by multiplication with $(1 + x^{n/2})$; the decoder works on 16-coefficient blocks consisting of the i -th octet and the octet at distance $n/2$. The specification bounds the DFR by (i) the inscribed ball $\|\text{Err}^{(i)}\|_2^2 < (q - 1)^2/2$ of the code and (ii) modelling $\|\text{Err}^{(i)}\|^2$ as $\sigma^2\chi^2(16)$, and reports failure rates from $2^{-143.9}$ (Light) to $2^{-204.1}$ (DTRU-2048).

Cheetah. Cheetah [2] is a Module-LWE KEM over $\mathbb{Z}_{7681}[x]/(x^{640} + 1)$ with rank $k \in \{1, 2, 3, 4\}$, centred binomial noise $\mathcal{B}_\eta$, one message bit per coefficient, and compression of the public key (d_b bits), of the first ciphertext component (d_u) and of the second (d_v). Its specification bounds the DFR by a Gaussian model of $e \cdot r + s \cdot e_1 - s \cdot e''$ against a threshold from which the maximum of the v -compression error has been subtracted, and reports 2^{-129} , 2^{-176} , 2^{-189} , 2^{-243} .

Rudraksh2 and Scabbard. Rudraksh2 [3] (Module-LWE with Kyber-style compression) and MORNING-Scabbard [4] (Module-LWR with power-of-two moduli) both use small rings ($n = 64$ at the 128-bit level) and encode four message bits into the coefficient pair $(i, i + n/2)$ with a two-dimensional B2-Minal code [5]. Their DFR computations evaluate the joint distribution of the pair by FFT and report 2^{-100} and 2^{-99} at the 128-bit level.

Contributions.

1. **DTRU: the failure event.** We derive the exact failure region of Algorithm 2. Because the two symbol values 0 and $(q+1)/2$ are antipodal in $\mathbb{Z}_q$, block i is decoded incorrectly iff for some codeword difference with support D ($|D| \in \{8, 16\}$) $\sum_{j \in D} |\text{Err}_j| \geq |D|(q+1)/4$. This is a union of half-spaces tangent to the inscribed ball, so the ball is a strict over-approximation.
2. **DTRU: the correlation.** In $x^n - x^{n/2} + 1$ every cross term of a ring product pairs indices at distance exactly $n/2$, and the repetition code makes $w_i = w_{i+n/2}$. Each decoding block is made of eight such pairs. The conditional covariance of a block, given the ciphertext, has eight eigenvalues around $0.3\times$ and eight around $1.6\text{--}2.1\times$ the marginal variance. The failing directions are the large ones.
3. **DTRU: a validated estimator and corrected DFRs.** We compute the Gaussian half-space union per block given the ciphertext randomness, correct the dominant terms with the exact centred-binomial cumulant generating function, and validate against the real scheme on scaled-down instances of all three rings. For 200 honest ciphertexts per set the average failure probabilities of DTRU-648/768/1024/Prime/1536/2048 are $2^{-85}, 2^{-98}, 2^{-105}, 2^{-96}, 2^{-101}, 2^{-111}$ against claimed values of 2^{-147} to 2^{-204} . One failure costs $2^{85}\text{--}2^{111}$ decapsulations; under a 2^{64} query cap the offline part is $2^{22}\text{--}2^{26}$ encapsulations for DTRU-648, -768 and -Prime.
4. **Cheetah: the noise expression and its exact tail.** Expanding Algorithms 15–17 of the specification, the decryption noise contains the public-key compression term $\langle e_b, r \rangle$, absent from the specification’s bound, and the v -compression error is a known, ciphertext-dependent shift of half-width $2^{\lceil \log_2 q \rceil - d_v - 1}$ rather than a worst case to subtract. We evaluate the tail exactly by exponential tilting and a one-dimensional FFT, validate the model on a scaled-down instance of the real algorithm (3 failures in 20000 encryptions, $2^{-12.70}$, against the exact $2^{-11.76}$) and on full-size noise (standard deviation to 1%, the $|E| \geq 800$ tail to 0.2 bits). The DFR of Cheetah128/256/384/512 is $2^{-79.1}, 2^{-51.8}, 2^{-116.1}, 2^{-201.0}$. Evaluating the specification’s own formula as written gives $2^{-163}, 2^{-126}, 2^{-149}, 2^{-336}$.
5. **Minal-coded schemes: the pair distribution.** Over $x^n + 1$ the noise pair $(\Delta m_0, \Delta m_{n/2})$ is a sum of $\ell n/2$ independent two-dimensional atoms $(xy + x'y', xy' - x'y)$. We evaluate its tail by one-dimensional projections onto the Voronoi radii of the real two-phase Minal decoder (phase 2 uses the threshold $q/4$), with a Mills/Rayleigh correction from the half-space to the radial tail, and validate on scaled-down instances of both schemes. The exact DFR is $2^{-103.9}$ for Rudraksh2-128-I and $2^{-102.1}$ for Scabbard-128, a few bits below the claimed 2^{-100} and 2^{-99} . The remaining Rudraksh2 and Scabbard sets claim DFRs well below 2^{-128} and are outside this paper.
6. **Failure boosting with a query cap.** We separate the two levers an encapsulator has: the squared norm of the ciphertext secret, whose cost is a large deviation of ℓn or kN binomial squares, and the ciphertext-known shift of each decoding position, which is free to inspect. For every set we report the cheapest way to bring a first failure within 2^{64} and 2^{80} decapsulation queries. DTRU-648 and Cheetah128/256 need no squared-norm tilt at 2^{80} ; Rudraksh2-128-I needs a tilt of $S = 1.34$ costing 2^{32} encapsulations and lands at a classical total of 2^{112} .

Every number in this paper is reproducible from a few hundred lines of Python; scripts and raw outputs accompany the paper.

2 Preliminaries

2.1 Notation

For a polynomial $a \in \mathcal{R} = \mathbb{Z}[x]/(\phi(x))$ of degree n we write a_j for its coefficients and $M_a \in \mathbb{Z}^{n \times n}$ for the matrix of multiplication by a , whose j -th column is $a \cdot x^j$; then $M_a b = ab$. For an index set I of size 16 we write $M_a[I]$ for the $16 \times n$ submatrix of rows I . $\mathcal{B}_\eta$ is the centred binomial distribution with variance $\eta/2$. $Q(z) = \Pr[N(0, 1) \geq z]$. $\text{mod}^\pm q$ denotes the representative in $(-q/2, q/2]$.

2.2 Failure boosting and the query cap

Following [6, 8], let $p(c)$ be the failure probability of an honestly generated ciphertext c over the secret key, and let $\delta = \mathbb{E}_c[p(c)]$. The adversary fixes a threshold f_t , generates ciphertexts offline (this is legal: the randomness is $H(\text{ID}(pk), m)$ or $G(m \| H(pk))$ for a chosen m), and keeps those with $p(c) > f_t$. With $\alpha = \Pr_c[p(c) > f_t]$ and $\beta = \mathbb{E}[p(c) \mid p(c) > f_t]$, the expected cost of one failure is α^{-1} encapsulations offline and β^{-1} decapsulation queries, i.e. total classical work $(\alpha\beta)^{-1}$; Grover search reduces the offline part to $\alpha^{-1/2}$. Since

$$\alpha\beta = \mathbb{E}[p(c) \mathbf{1}_{p(c) > f_t}] \leq \mathbb{E}[p(c)] = \delta, \quad (1)$$

classical total work never drops below $1/\delta$; boosting trades oracle queries for offline work. NIST caps the decapsulation oracle at 2^{64} queries; several NGCC submissions argue with 2^{80} . We report both: f_t is chosen so that β^{-1} equals the cap, and the total is $\alpha^{-1} \cdot 2^{\text{cap}}$. When δ itself exceeds $2^{-\text{cap}}$, random ciphertexts suffice and the total is $1/\delta$. Because the public key enters the hash, weak ciphertexts do not transfer between keys and the multi-target improvements of [8] do not apply.

What the encapsulator knows. $p(c)$ conditions on everything the encapsulator computes: the ciphertext secret vectors and every compression error of the ciphertext. The decryption noise at a decoding position therefore splits into a part that depends on the unknown key, whose distribution is set by the ciphertext, and a known shift d (for instance e_2 plus the v -compression error in Cheetah, or e'' plus the v -compression error in Rudraksh2). Two levers follow. First, the squared norm of the ciphertext secret scales the variance of the unknown part; a coefficient distribution with mean square S times its honest value has probability $2^{-\text{cost}(S)}$ given by the large deviation of ℓn (or kN) independent squares, computed from the exact cumulant generating function. Its ceiling $S_{\max}$ is the largest square over the honest mean square (4 for $\mathcal{B}_2$, 10 for $\mathcal{B}_5$). Second, the shift d at each of the n_{pos} decoding positions is inspected at no cost; selecting a position whose shift points towards the nearest failure boundary raises that position's failure probability, while the other $n_{\text{pos}} - 1$ positions keep their average. We optimise the two levers jointly.

2.3 Exact tails by exponential tilting

Every noise term in this paper is a sum of independent, bounded, exactly enumerable atoms (products of centred binomials, of a centred binomial and a rounding error, and so on). For a one-dimensional tail we tilt each atom by $e^{\theta x}/K(\theta)$ so that the tilted mean lies at the threshold, convolve the tilted atoms on a grid by FFT in double precision, and undo the tilt: $\Pr[X = z] = K(\theta)^N e^{-\theta z} p_\theta^{*N}(z)$. The Radon–Nikodym factor is applied only in a window around the tilted mode, so FFT floor noise is not reweighted into the far tail. For the two-dimensional Minal cell we project the same atoms onto each Voronoi radius of the real decoder and apply a Mills/Rayleigh factor $z\sqrt{2\pi}$ that converts the half-space tail into a radial tail; the construction

Set	ring	p	n	q	q_2	$(k_g, k_{f'})$	(k_r, k_e)	claimed $\log_2 \delta$	level
Light	pow2	$1 - x^{n/2}$	512	769	2^8	(2, 1)	(1, 2)	-143.92	128
648	tri	2	648	3457	2^9	(9, 2)	(2, 2)	-146.79	128
768	tri	2	768	3457	2^{10}	(4, 3)	(4, 2)	-185.56	192
1024	tri	2	1024	3457	2^{10}	(5, 2)	(3, 2)	-190.48	256
Prime	lppnf	2	1087	2017	2^{10}	(2, 1)	(2, 1)	-180.72	256
1536	tri	2	1536	3457	2^{10}	(3, 2)	(2, 1)	-195.50	384
2048	tri	2	2048	3457	2^{10}	(5, 1)	(2, 1)	-204.10	512

Table 1: DTRU parameter sets. DTRU-648 targets 128-bit quantum security (its NTRU Core-SVP claim is 164/144).

is exact for an isotropic law and a circular cell, and the scaled-down instances of Section 14 check the residual.

Part I

DTRU

3 The scheme

Three rings are used:

$$\mathcal{R}^{\text{tri}} = \mathbb{Z}[x]/(x^n - x^{n/2} + 1), \quad \mathcal{R}^{\text{pow2}} = \mathbb{Z}[x]/(x^n + 1), \quad \mathcal{R}^{\text{lppnf}} = \mathbb{Z}[x]/(x^n - x - 1).$$

Key generation samples $f' \leftarrow \mathcal{B}_{k_{f'}}^n$, $g \leftarrow \mathcal{B}_{k_g}^n$, sets $f = pf' + 1$ with $p = 2$ (tricyclotomic, LPPNF) or $p = 1 - x^{n/2}$ (power-of-two), and publishes $h = g/f \bmod q$. Encryption samples $r \leftarrow \mathcal{B}_{k_r}^n$, $e \leftarrow \mathcal{B}_{k_e}^n$, encodes the message to $w \in \{0, 1\}^n$ and outputs $c = \lfloor \frac{q_2}{q}(hr + e + \frac{q+1}{2}w) \rfloor \bmod q_2$. Decryption decompresses $c' = \lfloor \frac{q}{q_2}c \rfloor$, computes $c'f$ and decodes. Writing ε for the compression noise and $u = e + \varepsilon$, the specification's Theorem 1 gives

$$c'f = \text{Err} + \frac{q+1}{2}w, \quad \text{Err} = \begin{cases} gr + f'(2u + w) + u, & p = 2, \\ gr + f'((1 - x^{n/2})u + w') + u, & p = 1 - x^{n/2}. \end{cases} \quad (2)$$

DoubleE₈. Each 4-bit chunk m_i is mapped to $u_i = m_i H \bmod 2 \in \{0, 1\}^8$ with H the generator of the $[8, 4, 4]$ extended Hamming code, placed at coefficients $8i, \dots, 8i + 7$ of w' , and $w = (1 + x^{\text{off}})w'$ with $\text{off} = n/2$. Hence $w_j = w_{j+n/2} = w'_j$. Algorithm 2 decodes block i from the 16 coefficients $t_{8i}, \dots, t_{8i+7}, t_{8i+n/2}, \dots, t_{8i+7+n/2}$ by computing, for each of the 16 codewords $\frac{q+1}{2}(u, u)$, the sum of squared centred distances $\|\cdot\|_{q,2}^2$ and returning the argmin. For $n = 1087$ (DTRU-Prime) the specification does not state the layout; we use $\lfloor n/16 \rfloor = 67$ blocks and $\text{off} = \lfloor n/2 \rfloor = 543$.

Parameters. Table 1 lists the seven sets (specification Table 1 and 2). The KEM is $\text{FO}_{\text{ID}(pk),m}^\chi$ with implicit rejection and $(K, \rho) = H(\text{ID}(pk), m)$; the encryption randomness is derived from ρ .

The specification's DFR bound. Section 2.3 of [1] argues that block i decodes correctly whenever

$$\|\text{Err}^{(i)}\|_2^2 = \|\text{Err}_i\|_{q,2}^2 + \|\text{Err}_{i+n/16}\|_{q,2}^2 < \frac{(q-1)^2}{2} =: T, \quad (3)$$

because the scaled code has minimum Euclidean distance $\sqrt{2}(q-1)$. It then assumes the 16 coefficients independent, so that $\|\text{Err}^{(i)}\|^2 \sim \sigma^2 \chi^2(16)$, and applies a union bound over $n/16$ blocks: $\delta \leq \frac{n}{16} \Pr[\chi^2(16) \geq T/\sigma^2]$. Section 2.4 adds that adversarial choice of the message can raise the failure rate by at most about 2^{20} .

4 The failure region of Algorithm 2

Lemma 1. *Let $u \in \{0,1\}^8$ be the sent codeword and let all coefficients of $\text{Err}^{(i)}$ be represented in $(-q/2, q/2)$. Algorithm 2 returns a codeword $u' \neq u$ only if, with $D = \text{supp}(u' - u) \cup (\text{supp}(u' - u) + 8)$,*

$$\sum_{j \in D} |\text{Err}_j^{(i)}| \geq |D| \cdot \frac{q+1}{4}, \quad (4)$$

and conversely it fails whenever (4) holds for some codeword $u' \neq u$ (up to rounding of ± 1 in the threshold).

Proof. Let $h = (q+1)/2$. On a coordinate where $u_j \neq u'_j$, the received value is $hu_j + e_j$ and the two candidate symbols are hu_j and $hu_j + h \equiv hu_j - (q-1)/2$. The centred distance to the wrong symbol is $h - |e_j|$ (up to one unit) regardless of the sign of e_j , because 0 and h are antipodal in $\mathbb{Z}_q$. The cost difference of u' against u is therefore $\sum_{j \in D} ((h - |e_j|)^2 - e_j^2) = \sum_{j \in D} (h^2 - 2h|e_j|)$, which is negative iff $\sum_{j \in D} |e_j| > |D|h/2$. The codeword pair is repeated on both octets, so D contains a position and its partner at distance 8 inside the block. $\square$

The $[8, 4, 4]$ code has 14 codewords of weight 4 and one of weight 8, so a block has 14 half-space families with $|D| = 8$ (threshold $2(q+1)$) and one with $|D| = 16$. Writing (4) as $\max_{s \in \{\pm 1\}^D} \langle s, \text{Err}^{(i)} \rangle \geq |D|(q+1)/4$, the failure region is the union over $14 \cdot 2^8 + 2^{16}$ half-spaces, each at Euclidean distance $(q+1)/\sqrt{2}$ from the origin. The ball (3) has radius $(q-1)/\sqrt{2}$: it is the inscribed ball of the Voronoi cell and a strict subset of the correct-decoding region. We verified Lemma 1 against the reference decoder on 8000 random error vectors at $q \in \{193, 3457\}$: the predicate never fired on a correctly decoded block and missed only a handful of failures at the modular seam.

A *fixed-sign* variant that only keeps $s = u' - u$ misses almost all failures (2–4 of several hundred in our tests): because of the antipodal symbols, both signs contribute. This matters below, since the fixed-sign direction is the low-variance direction of the block.

5 Block covariance in the tricyclotomic ring

Fix the ciphertext randomness (r, u, w) and set $v = 2u + w$ (tricyclotomic and LPPNF) or $v = (1 - x^{n/2})u + w'$ (power-of-two). By (2), $\text{Err} = M_r g + M_v f' + u$ is linear in the secret (g, f') , whose coefficients are i.i.d. centred binomial with variances $\sigma_g^2 = k_g/2$ and $\sigma_{f'}^2 = k_{f'}/2$. For a block with index set I ,

$$\Sigma_I = \sigma_g^2 M_r[I] M_r[I]^\top + \sigma_{f'}^2 M_v[I] M_v[I]^\top = \sigma_g^2 \text{Gram}_I(r) + \sigma_{f'}^2 \text{Gram}_I(v). \quad (5)$$

The specification's model corresponds to replacing Σ_I by its diagonal (in fact by a single scalar).

Lemma 2. *In $\mathbb{Z}[x]/(x^n - x^{n/2} + 1)$, for every $a \in \mathcal{R}$ and every output index k , each coefficient a_j appears in $(a \cdot x^t)_k$ for exactly one t except for pairs (j, j') with $|j - j'| = n/2$, which appear together with equal sign. Consequently $\text{Gram}(a)_{k, k+n/2} \neq 0$ in general, while for $|k - k'| \notin \{0, n/2\}$ the entry $\text{Gram}(a)_{k, k'}$ has mean zero over random a .*

The proof is a direct computation with $x^n = x^{n/2} - 1$; in the power-of-two ring $x^n = -1$ produces no such pairing, and in the LPPNF ring $x^n = x + 1$ produces an analogous but weaker coupling. The consequence for DTRU is that a decoding block consists of exactly the eight ring-paired index pairs $(8i + j, 8i + j + n/2)$, $j = 0, \dots, 7$, and the encoding repetition $w_j = w_{j+n/2}$ adds a second correlated component through $\text{Gram}(v)$.

Numerical picture (DTRU-2048). Sampling the secret 3000 times for one honest ciphertext, coefficients 0 and $n/2$ of Err have variances $1.09 \cdot 10^4$ and $1.13 \cdot 10^4$ and covariance $-7.3 \cdot 10^3$ (correlation -0.66). The expected block covariance of block 0 has eigenvalues

$$\{5709, 5723, 5781, 5788, 5797, 5843, 5851, 6767\} \cup \\ \{28964, 28993, 29110, 29394, 29537, 29557, 29580, 38418\},$$

against a marginal variance of about $1.84 \cdot 10^4$. The trace is preserved: correlation redistributes variance, it does not create it. Since the failure region is a union of half-spaces, the tail is governed by $\max_s s^\top \Sigma_I s$, i.e. by the large eigenvalues, and the sign-alternating patterns s with $s_j = -s_{j+8}$ are exactly the ones aligned with them. Across the parameter sets the ratio between the largest eigenvalue and the mean diagonal is 2.1–2.5 (tricyclotomic), 3.4 (LPPNF) and 1.27 (power-of-two, DTRU-Light), see Table 3.

6 Estimating the failure probability

Per-ciphertext probability. Given (r, u, v) , for each block I and each pattern (D, s) we compute $\sigma_s^2 = s^\top \Sigma_I s$ and the Gaussian tail $Q(|D|(q+1)/4 \mp \langle s, u_I \rangle) / \sigma_s$ for both signs of s (the known term u shifts the mean); the block probability is the union bound over patterns, and $p(c)$ the union over blocks. At the operating points of DTRU the dominant pattern has standard score $z \approx 12$ –13, the half-spaces barely overlap, and the union bound is tight; on the toy parameters below (with $z \approx 3$ –4) it overestimates the mean by 1–2 bits, which we report.

Exact secret distribution. The Gaussian approximation of $\langle s, \text{Err} \rangle = a_g^\top g + a_f^\top f'$ at 12σ deserves a check. The exact cumulant generating function is $K(t) = \sum_j 2k_g \log \cosh(ta_{g,j}/2) + \sum_j 2k_{f'} \log \cosh(ta_{f,j}/2)$, and the Lugannani–Rice saddlepoint formula gives the tail. Applied to the 64 most probable (I, D, s) terms of each ciphertext, it lowers $\log_2 p$ by 0.4–1.3 bits; all headline numbers include this correction.

Averaging over ciphertexts. For each set we draw 200 honest ciphertexts and report the median of $\log_2 p(c)$, the sample mean $\bar{p}$ of $p(c)$ (with a bootstrap interval), and the normal-tail extrapolation $\log_2 \mathbb{E}[p] = \mu + \sigma^2 \ln 2 / 2$ from the sample mean μ and standard deviation σ of $\log_2 p$. The sample mean is dominated by the few weakest ciphertexts; the extrapolation places the mass of $\mathbb{E}[p]$ at $\mu + \sigma^2 \ln 2$, i.e. 4–7 σ above the mean, beyond the sample. Within the sample, $\log_2 p$ is close to normal (skewness -0.23 to $+0.04$, excess kurtosis about 0). Since $\log_2 p$ is a concave function of the top eigenvalue, a Gaussian-tailed eigenvalue would make the true right tail lighter than normal, so the truth is expected between the two estimates.

6.1 Validation on scaled-down instances

For each ring we ran the complete scheme (key generation with polynomial inversion, compression, decompression, Algorithm 2) on 40 fresh keys with 500 encryptions each, keeping the CBD widths of a real parameter set and lowering (n, q) until failures become observable. Table 2 compares the observed failure rate with the ball condition (3) and with the model of this section evaluated on the same ciphertexts. The algebraic error (2) matched the integer computation of $c'f$ coefficient by coefficient in all 60000 encryptions.

Ring	toy parameters	Alg. 2 fails	ball	model median	model mean	$\mathbb{E}[\text{Err}^2]$ ratio
$x^n - x^{n/2} + 1$	$n=64, q=193, q_2=64, \mathcal{B}_{5,1,2,1}$	$2^{-9.7}$ (24)	$2^{-4.7}$	$2^{-10.0}$	$2^{-8.1}$	1.03
$x^n + 1, p=1-x^{32}$	$n=64, q=89, q_2=32, \mathcal{B}_{2,1,1,2}$	$2^{-8.1}$ (71)	$2^{-2.2}$	$2^{-8.4}$	$2^{-6.9}$	1.02
$x^n - x - 1$	$n=61, q=149, q_2=64, \mathcal{B}_{2,1,2,1}$	$2^{-8.4}$ (58)	$2^{-3.8}$	$2^{-9.2}$	$2^{-6.6}$	1.01

Table 2: Real decoder versus models on scaled-down instances (20000 encryptions each). “ball” is the rate of $\|\text{Err}^{(i)}\|^2 \geq T$; the last column is the empirical second moment of the error coefficients over the model’s $\text{tr } \Sigma_I/16$.

Set	claimed	indep. χ^2 + ball	σ^2 ours/ implied	spectrum + ball, mean	median	true decoding failure mean [95%]	normal tail	$\lambda_{\max}/$ diag
Light	-143.9	-139.2	1177/1145	-101.4	-153.6	-137.4 [-139.9, -136.2]	-121.7	1.27
648	-146.8	-153.3	21978/22777	-70.4	-96.8	-85.4 [-89.9, -84.1]	-83.6	2.13
768	-185.6	-189.7	18377/18719	-74.9	-111.6	-97.9 [-100.5, -96.6]	-86.0	2.31
1024	-190.5	-196.7	17783/18276	-83.9	-117.6	-104.6 [-108.2, -103.4]	-98.1	2.25
Prime	-180.7	-224.7	5402/6498	-64.4	-115.2	-96.3 [-102.0, -94.7]	-79.8	3.43
1536	-195.5	-195.6	17826/17834	-77.3	-110.7	-100.7 [-102.9, -99.4]	-93.9	2.50
2048	-204.1	-209.6	16781/17166	-88.6	-123.2	-110.8 [-115.2, -109.3]	-109.1	2.32

Table 3: $\log_2$ of failure probabilities, 200 honest ciphertexts per set. “implied” is the per-coefficient variance that reproduces the claimed DFR under the specification’s model.

The ball condition fires 4–6 bits more often than the decoder fails. The per-ciphertext model matches the decoder to 0.3–0.8 bits at the median; its mean is inflated by the union bound at these moderate standard scores. Per-key failure counts ranged from 0 to 21, so the failure rate of a fixed key deviates from the average by a few bits.

7 Corrected failure rates

Table 3 gives, for each set: the claimed DFR; the specification’s method (independent $\chi^2(16)$ on the ball) re-run with our marginal variance; the ball event with the correct spectrum; and the true decoding failure with the correct spectrum (median, sample mean with a 95% bootstrap interval, normal-tail extrapolation).

Three observations.

1. The third column reproduces the specification’s figures to within ± 8 bits for all sets except DTRU-Prime, where the specification’s implied variance is 20% larger than ours (its independent model would give 2^{-225} rather than the claimed 2^{-181}). The discrepancy between claimed and true DFR is therefore a modelling error, not a variance error.
2. For the six sets over the tricyclotomic and LPPNF rings the true failure probability is 50–110 bits larger than claimed. The independence assumption alone accounts for about 100 bits (the true decoding event under $\Sigma_I = \lambda I$ would be 2^{-221} for DTRU-2048); the ball over-approximation goes the other way by 18–50 bits. The specification’s 2^{20} allowance for adversarial messages does not cover this.
3. DTRU-Light over $x^n + 1$ has no ring pairing ($\lambda_{\max}$ is only $1.27\times$ the diagonal) and its median DFR is below the claim. Its spread across ciphertexts is however the largest ($\sigma = 9.6$ bits at $n = 512$), and the two estimates of $\mathbb{E}[p]$, 2^{-137} and 2^{-122} , straddle the 128-bit level.

8 Failure boosting against DTRU

Set	level	Core-SVP ours/claimed	$\log_2$ work (sample mean)	(normal tail)	verdict (DFR)
Light	128	132.3 / 132	137.4	121.7	borderline
648	128	164.7 / 164	85.4	83.6	−43 bits
768	192	199.4 / 195	97.9	86.0	−94 to −106
1024	256	277.7 / 270	104.6	98.1	−151 to −158
Prime	256	292.6 / 280	96.3	79.8	−160 to −176
1536	384	416.7 / 411	100.7	93.9	−283 to −290
2048	512	512.5 / 567	110.8	109.1	−401

Table 4: Classical work for one decryption failure (decapsulations), without a query cap. Core-SVP values are for the NTRU key-recovery problem; the DTRU-2048 estimate is a primal-BDD fallback with low confidence.

Set	$f_t = 2^{-80}$ (oracle within 2^{80})				oracle capped at 2^{64}			
	$\#\{p > 2^{-80}\}/200$	precomp	oracle	classical	precomp	oracle	classical	quantum
648	2	8.3	75.4	83.7	24.1	62.2	86.3	74.2
768	0	13.0	73.0	86.0	25.9	60.6	86.6	73.6
Prime	0	12.1	67.7	79.8	22.3	57.5	79.8	68.7
1024	0	21.8	77.1	98.9	41.1	62.4	103.4	82.9
1536	0	17.6	76.9	94.5	36.6	62.4	99.0	80.7
2048	0	37.4	78.7	116.1	67.3	63.1	130.4	96.8
Light	0	46.8	77.7	124.5	67.5	62.3	129.8	96.1

Table 5: Offline precomputation ($\log_2 \alpha^{-1}$ encapsulations) and oracle queries ($\log_2 \beta^{-1}$) under the normal-tail fit. For DTRU-648 the 2^{-80} threshold is inside the sample: two of 200 honest ciphertexts have $p > 2^{-80}$ (weakest $2^{-78.7}$). The 2^{64} -cap entries for DTRU-2048 and DTRU-Light are 9σ extrapolations.

8.1 Cost of the first failure

By (1) the classical total work for one failure is at least $1/\mathbb{E}[p]$. Table 4 lists it for the sample-mean and the normal-tail estimates, together with the nominal level and the NTRU Core-SVP estimates (ours, from the lattice estimator, and the designers’).

Filtering within the sample (keeping the weakest quarter, or the single weakest ciphertext) leaves the classical total unchanged and saves 3–4 bits quantumly, as (1) predicts. We also tried tilting the distributions of r, e and of the codeword weight (exponential families on the expected covariance): on DTRU-2048 this gains only a few bits, because $\|r\|^2$ is concentrated at $n = 2048$ and the weak ciphertexts are those with a large top eigenvalue (correlation 0.99 between $\log_2 p$ and the top eigenvalue on the ball statistic), not those with a large norm.

8.2 Precomputation versus oracle queries

Following [6], the offline and online costs separate. We fit $\log_2 p$ by a normal law and ask for the threshold f_t at which the conditional failure probability is 2^{-80} (the oracle then stays within a 2^{80} cap), and then for the threshold at which $\beta^{-1} = 2^{64}$.

For DTRU-648, -768 and -Prime the first failure fits into a 2^{64} query budget after 2^{22} – 2^{26} offline encapsulations; for DTRU-1024 and -1536 the offline part is 2^{37} – 2^{41} . The total work remains 2^{80} – 2^{130} , so none of this is a practical attack; the point is that the designers’ query-limit argument (“ $2^{64} \cdot 2^{-147}$ is negligible”) does not hold once weak ciphertexts are selected offline.

9 After the first failure

A failing ciphertext with its most probable pattern (I, D, s) tells the adversary that $\langle s, \text{Err}_I \rangle = a_g^\top g + a_f^\top f' + \langle s, u_I \rangle$ exceeded $|D|(q+1)/4$, where $a_g = M_r[I]^\top s$ and $a_f = M_v[I]^\top s$ are known. Implicit rejection hides the block and the pattern, so the posterior is a mixture whose largest component has weight around 10%; the sign of $\langle a, x \rangle$ is not revealed either.

For DTRU-2048 (24 honest ciphertexts) the most probable pattern has standard score 12.97 (median), the direction a has participation ratio $(\sum a_i^2)^2 / \sum a_i^4 \approx 891$ out of 4096 coordinates, and directions of different ciphertexts have absolute cosine 0.19 (median) to 0.27 (max). Conditioning on a first failure along a_1 , a second ciphertext with direction a_2 at correlation ρ fails with standard score $z' = (z - \rho z) / \sqrt{1 - \rho^2}$, i.e. 10.7 ($\rho = 0.19$) to 9.8 ($\rho = 0.27$), so its failure probability is 2^{-86} to 2^{-72} instead of 2^{-121} . This is the mechanism of directional failure boosting [7]: after the first failure, further failures are cheap, and the total cost is that of the first one.

On the $n = 64$ toy instance we collected 150 real decoder failures for one key and estimated the secret up to sign by the top eigenvector of $\sum_k a_k a_k^\top$: the estimate has correlation 0.22 with the true secret (random directions: 0.07), with only $z \approx 3.5$ per failure. At the real parameters each failure carries $z \approx 13$. We did not run the final lattice-reduction step and do not claim a completed key recovery.

Part II

Cheetah

10 The scheme and its decryption noise

Cheetah works in $\mathcal{R}_q = \mathbb{Z}_q[x]/(x^N + 1)$ with $q = 7681$, $N = 640$ and module rank k . With $d_q = \lceil \log_2 q \rceil = 13$, compression to d bits is $\text{Compress}(x, D) = \lfloor x/D \rfloor$ and $\text{Decompress}(y, D) = \min(yD + D/2, q - 1)$ for $D = 2^{d_q - d}$; the error $x' - x$ takes the values $D/2 - (x \bmod D)$, so it is uniform on $\{-D/2 + 1, \dots, D/2\}$ with mean $1/2$. Key generation samples $s, e \leftarrow \mathcal{B}_\eta^{k \times N}$ and publishes $\tilde{b} = \text{Compress}(As + e, 2^{d_q - d_b})$. Encryption decompresses $b = As + e + e_b$, samples $r, e_1 \leftarrow \mathcal{B}_\eta^{k \times N}$, $e_2 \leftarrow \mathcal{B}_\eta^N$, and outputs $u = \text{Compress}(r^\top A + e_1, 2^{d_q - d_u})$ and $v = \text{Compress}(\text{Truncate}(r^\top b + e_2, \lambda) + \Delta m, 2^{d_q - d_v})$ with $\Delta = 2^{-1} \bmod q$; only the first λ coefficients of v are sent. Decryption computes $m' = (2(v' - \text{Truncate}(u'^\top s, \lambda)) \bmod \pm q) \bmod 2$. Writing $\Delta u, \Delta v$ for the two ciphertext compression errors,

$$v' - u'^\top s = \Delta m + E, \quad E = \langle e, r \rangle + \langle e_b, r \rangle - \langle s, e_1 \rangle - \langle s, \Delta u \rangle + e_2 + \Delta v, \quad (6)$$

and since $2\Delta \equiv 1$, coefficient j decodes correctly iff $|2E_j + m_j| < q/2$, i.e. (up to one unit) $|E_j| < q/4$.

The specification's bound. Section 2.2 of [2] writes the noise as $e \cdot r + e_2 + e' - s \cdot e_1 - s \cdot e''$: the public-key term $\langle e_b, r \rangle$ is absent, although the encryptor multiplies r by the decompressed b (Algorithm 16, lines 4–5 and 15). It then subtracts the maximum $2^{d_q - d_v - 1}$ of $e' = \Delta v$ and the bound η on $e'' = \Delta u$ from the threshold, models the remaining terms as a Gaussian of standard deviation $\sqrt{3n\sigma^2}$, and reports Table 6. For Cheetah128 and Cheetah256 ($d_b = 10$, $D_b = 8$) the omitted term has second moment $kN \cdot 5.5 \cdot \eta/2$, more than twice that of $\langle e, r \rangle$ itself; for $d_v = 4$ the subtracted maximum is 256 out of a threshold of 1920, while the actual Δv is a known shift that is as often favourable as not.

Set	k	λ	η	(d_b, d_u, d_v)	claimed $\log_2 \delta$	primal C/Q	Core-SVP C/Q
Cheetah128	1	128	5	(10, 10, 4)	-129	159 / 139	148.3 / 134.6
Cheetah256	2	256	4	(10, 10, 4)	-176	307 / 276	325.9 / 295.7
Cheetah384	3	384	3	(11, 11, 4)	-189	426 / 382	499.0 / 452.9
Cheetah512	4	512	2	(11, 11, 8)	-243	541 / 497	511.9 / 464.6

Table 6: Cheetah parameter sets (specification Table 2) with the designers’ primal-attack estimates (Table 4) and the Core-SVP values of the lattice estimator (0.292β / 0.265β).

Set	level	claimed	§2.2 formula	exact $\log_2 \delta$	primal / Core-SVP	2^{80} queries		2^{64} queries	
						S , pre	C / Q	S , pre	C / Q
Cheetah128	128	-129	-162.8	- 79.1	159 / 148.3	1, 0	79.1 / 79.1	1.47, 49.4	113.4 / 88.7
Cheetah256	256	-176	-125.5	- 51.8	307 / 325.9	1, 0	51.8 / 51.8	1, 0	51.8 / 51.8
Cheetah384	384	-189	-149.2	-116.1	426 / 499.0	1.83, 549	629 / 355	2.67, 1386	1450 / 757
Cheetah512	512	-243	-335.6	-201.0	541 / 511.9	3.5, 6422	6502 / 3291	4.0, 15360	15424 / 7744

Table 7: Cheetah: the specification’s claimed DFR, the same §2.2 formula evaluated as written, the exact tilted-FFT DFR, the designers’ primal estimate and the classical Core-SVP value, and the cheapest first failure under each query cap. S is the squared-norm tilt of r (and, when helpful, e_1) and “pre” its $\log_2$ offline cost; C / Q are the classical and quantum totals.

11 Exact tail and validation

Unknown part and known shift. From the encapsulator’s point of view E_j splits into the shift $d_j = e_{2,j} + \Delta v_j$, known exactly, and the unknown part $Z_j = \langle e + e_b, r \rangle_j - \langle s, e_1 + \Delta u \rangle_j$, a sum of kN atoms $(e + e_b)_a r_b$ and kN atoms $s_a (e_1 + \Delta u)_b$, each enumerated exactly (the atom pmfs have at most a few hundred points). The second moment of the first atom is $\mathbb{E}[(e + e_b)^2] \mathbb{E}[r^2]$, which includes the mean $1/2$ of e_b . We compute the two-sided tail $\Pr[|E| \geq q/4]$ by the tilted FFT of Section 2.3 on a grid of at most 2^{16} points, folding e_2 and Δv into the same convolution; then $\delta = 1 - (1 - p)^\lambda$ with p that coefficient tail. A squared-norm tilt of r and e_1 rebuilds the ciphertext-side atoms so that $\mathbb{E}[r_j^2] = S_r \cdot \eta/2$ and likewise for e_1 , and the offline cost is the large-deviation probability of those squares.

Scaled-down instance. We implemented Algorithms 15–17 literally (negacyclic multiplication, $\text{Compress}(x, D) = \lfloor x/D \rfloor$, $\text{Decompress} = \min(yD + D/2, q - 1)$, the mod2 decoder) at $N = 64$, $k = 2$, $q = 769$, $\eta = 4$, $(d_b, d_u, d_v) = (8, 8, 5)$, $\lambda = 32$, on 40 keys with 500 encryptions each. The decoder failed on 3 of 20000 ciphertexts ($2^{-12.70}$); the exact computation above gives $2^{-11.76}$. The predicate $|E_j| \geq q/4$ agreed with the decoder on every ciphertext. The empirical second moment of E is 1.052 of the sum of the six exact terms.

Full-size noise. At the real parameters we sampled E on the transmitted positions of Cheetah128 (800 encryptions) and Cheetah256 (1200 encryptions). The empirical standard deviation is 217.0 against the model 216.6 for Cheetah128 and 245.8 against 243.7 for Cheetah256. The empirical frequency of $|E| \geq 800$ is $2^{-14.06}$ against the exact $2^{-14.16}$ (Cheetah128) and $2^{-10.76}$ against $2^{-10.63}$ (Cheetah256). No coefficient reached 1000 in these trials (the exact model puts that event at $2^{-22.2}$ and $2^{-16.2}$). No decryption failure occurred, as expected at 2^{-79} and 2^{-52} ; the value at $q/4 = 1920.25$ is the same tilted FFT continued to the threshold.

12 Results for Cheetah

Table 7 summarises. Cheetah256 fails with probability $2^{-51.8}$ on an honest ciphertext; random ciphertexts produce a failure after 2^{52} decapsulations, below the nominal 256, the designers’

primal estimate of 307 and the Core-SVP value of 325.9. Cheetah128 fails at $2^{-79.1}$; under a 2^{80} cap random ciphertexts suffice (2^{79} against a level of 128), and under a 2^{64} cap a squared-norm tilt of $S_r = 1.47$ with 2^{49} offline encapsulations gives a classical total of 2^{113} and a quantum total of 2^{89} . Cheetah384 and Cheetah512 are 73 and 42 bits above their claims but cannot be brought under either cap without an offline search far above the level; without a cap their first failure costs 2^{116} and 2^{201} , below the levels 384 and 512 and below every lattice estimate in Table 6. We keep those two sets out of the first-class list of Section 16 because the 2^{80} total exceeds the level.

The v -compression at $d_v = 4$ is the design decision behind the two low levels: the known shift has half-width 256 against a threshold of 1920 and a total noise of standard deviation 217–244, and the public-key compression at $d_b = 10$ adds a term of the same size as the LWE error itself. Cheetah512 uses $d_v = 8$ and $d_b = 11$ and is affected only by the e_b term and the exact (heavier than Gaussian) tail. The specification’s own formula, evaluated as written, is more conservative than the claim for Cheetah128 and Cheetah512 and less conservative for Cheetah256 and Cheetah384; neither the claim nor that formula includes $\langle e_b, r \rangle$.

Part III

Minal-coded schemes: Rudraksh2 and Scabbard

13 B2-Minal codes and the failure region

Both schemes encode four bits $m = (m_0, m_1, m_2, m_3)$ into the coefficient pair $(i, i + n/2)$ of a polynomial in $\mathbb{Z}_q[x]/(x^n + 1)$ as $c = G_2 m$ with

$$G_2 = \begin{pmatrix} \lfloor q/2 \rfloor & 0 & \lfloor q/4 \rfloor & \beta \\ 0 & \lfloor q/2 \rfloor & \beta & \lfloor q/4 \rfloor \end{pmatrix},$$

where $\beta < q/2$ is a tailoring parameter [5]. Decoding (Algorithm 6 of [4], Algorithm 7 of [3]) first reduces the pair modulo $\lfloor q/2 \rfloor$ and picks (m_2, m_3) by minimum centred Euclidean distance among the four points of the pure Minal code generated by $[\lfloor q/4 \rfloor, \beta]$; then (m_0, m_1) are the indicators $|\delta_x| \geq q/4$, $|\delta_y| \geq q/4$ of the centred residual against the chosen point. The correct-decoding region of a message is a polygon in $\mathbb{Z}_q^2$. We do not replace it by an inscribed ball: the failure radii of the vectorised decoder, averaged over the 16 messages, are the thresholds of Section 14.

Noise. For Rudraksh2 (Module-LWE, rank ℓ , $\mathcal{B}_\eta$ secrets, Kyber-style rounding compression to p and t bits)

$$\Delta m = \langle e, s' \rangle - \langle s, e' + \Delta u \rangle + e'' + \Delta v,$$

where s, e are the key, s', e', e'' the ciphertext secrets and $\Delta u, \Delta v$ the compression errors. For Scabbard (Module-LWR, moduli $q = 2^{\epsilon_q}$, $p = 2^{\epsilon_p}$, $t = 2^{\epsilon_t}$) the public key is $b = \lfloor (A^\top s + h) \rfloor_{\epsilon_q - \epsilon_p}$ and the ciphertext $u = \lfloor (As' + h) \rfloor_{\epsilon_q - \epsilon_p}$, $v = (2^{\epsilon_q - \epsilon_p} b^\top s' - \mu \bmod q) \gg (\epsilon_q - \epsilon_t - 2)$, and decryption forms $m' = 2^{\epsilon_q - \epsilon_p} u^\top s - 2^{\epsilon_q - \epsilon_t - 2} v \bmod q$ (Algorithms 9–11 of [4]). Writing e_b, e_u for the two rounding errors (uniform on $\{-2^{d-1} + 1, \dots, 2^{d-1}\}$ with $d = \epsilon_q - \epsilon_p$, thanks to the constant h) and ρ for the remainder discarded from v ,

$$m' = \mu + \Delta m, \quad \Delta m = \langle e_u, s \rangle - \langle e_b, s' \rangle + \rho.$$

Lines 8 of Algorithms 10 and 11 contain no rounding constant for v . We model ρ as centred on $\{-2^{\epsilon_q - \epsilon_t - 3}, \dots, 2^{\epsilon_q - \epsilon_t - 3} - 1\}$, the law a Saber-style constant would produce and the reading consistent with the claimed 2^{-99} . For Scabbard-128 the half-width is 256.

14 Exact joint distribution of a pair

Lemma 3. *Let $a, b \in \mathbb{Z}[x]/(x^n + 1)$ have i.i.d. symmetric coefficients from distributions A and B . The pair $((ab)_0, (ab)_{n/2})$ is distributed as the sum of $n/2$ independent copies of $(xy + x'y', xy' - x'y)$ with $x, x' \sim A$ and $y, y' \sim B$ independent.*

Proof. Coefficient 0 of ab uses the products $a_i b_{-i}$ and coefficient $n/2$ the products $a_i b_{n/2-i}$ (indices modulo n , with the sign of the wrap-around). For $i \in \{0, \dots, n/2 - 1\}$ the four variables $a_i, a_{i+n/2}, b_{-i}, b_{n/2-i}$ appear only in the two products of index i and the two of index $i + n/2$, and the index sets are disjoint across i . Collecting signs, the contribution of this quadruple is $(\pm a_i b_{-i} \pm a_{i+n/2} b_{n/2-i}, \pm a_i b_{n/2-i} \pm a_{i+n/2} b_{-i})$ with sign product -1 ; flipping the signs of the symmetric variables $a_{i+n/2}$ and b_{-i} brings it to the stated form. $\square$

The atom has uncorrelated coordinates of equal variance $\mathbb{E}[x^2]\mathbb{E}[y^2] \cdot 2$, but it is not Gaussian and not isotropic in its higher cumulants. For Rudraksh2 the pair $(\Delta m_0, \Delta m_{n/2})$ is the sum of $\ell n/2$ atoms with $A = \mathcal{B}_\eta$, $B = \mathcal{B}_\eta$ (from $\langle e, s' \rangle$), $\ell n/2$ atoms with $A = \mathcal{B}_\eta$, $B = \mathcal{B}_\eta * \Delta u$, and the known shift $d = (e'' + \Delta v)_{0, n/2}$. For Scabbard it is the sum of $\ell n/2$ atoms with A the rounding error and $B = \mathcal{B}_\eta$, twice, plus the shift ρ . All atom pmfs are enumerated exactly ($5^4 = 625$ terms for two $\mathcal{B}_2$ factors, $16^2 \cdot 5^2$ for Scabbard).

Computation. The B2-Minal cell is star-shaped. For each of the 16 messages and each of 32 directions ϕ we take the first-fail radius $r(\phi)$ of the vectorised decoder. The atoms of Lemma 3 project onto that direction as one-dimensional laws; their sum is evaluated by the same tilted FFT as in Section 2.3, and a Mills/Rayleigh factor $z\sqrt{2\pi}$ with $z = r/\sigma_{\text{proj}}$ converts the half-space tail into a radial tail. Averaging over messages and angles gives the pair failure probability; then $\delta = 1 - (1 - \delta_{\text{pair}})^{n/2}$. A squared-norm tilt of the ciphertext secrets rebuilds the atoms so that their coefficient squares have mean S times the honest value.

Validation. On scaled-down instances of both schemes we ran the real algorithms (matrix sampling, CBD, Kyber-style or power-of-two rounding, Minal encoding and decoding) and compared the ciphertext failure rate with the exact pair computation. For Rudraksh2 at $\ell = 2, n = 16, q = 97, p = 32, t = 16, \eta = 1, \beta = 8$ we observed 699 failures in 2000 encryptions ($2^{-1.52}$) against the exact $2^{-1.68}$; the predicate $|(\Delta m_0, \Delta m_{n/2})| \in F_m$ agreed with the decoder on every ciphertext, and the empirical coefficient variance is 0.972 of the model. For Scabbard at $\ell = 2, n = 16, \epsilon_q = 8, \epsilon_p = 5, \epsilon_t = 3, \eta = 2, \beta = 12$ the toy is past the high-noise regime: 1467 of 2000 encryptions failed and the exact computation saturates at 1, with second-moment ratio 1.031. These toys check the decoder and the second moment; the operating-point tails of Section 15 use the same atoms and radii at the real (q, β) .

15 Results for Rudraksh2-128-I and Scabbard-128

Rudraksh2-128-I ($\ell = 9, n = 64, q = 3329, p = 2^{12}, t = 2^6, \eta = 2, \beta = 220$). The exact DFR is $2^{-103.9}$, three to four bits below the claimed 2^{-100} . Under a 2^{80} query cap a squared-norm tilt of the ciphertext secrets to $S = 1.34$ makes $\delta(S) \approx 2^{-80}$ at an offline cost of 2^{32} encapsulations, for a classical total of 2^{112} and a quantum total of 2^{96} (Table 8). The classical total sits 16 bits below the nominal 128; the quantum total exceeds the 80-bit quantum target of the submission. Under a 2^{64} cap the tilt must reach $S = 1.67$, the offline cost is 2^{104} and the classical total 2^{168} (quantum 2^{116}). Without a cap a failure costs 2^{104} : below the nominal 128, below the designers' Core-SVP 129 and below the estimator's 197.7.

Set	level	claimed	exact $\log_2 \delta$	2^{80} queries		2^{64} queries	
				S , precomp	classical / quantum	S , precomp	classical / quantum
Rudraksh2-128-I	128 / 80q	-100	-103.9	1.34, 32.2	112.2 / 96.1	1.67, 103.8	167.8 / 115.9
Scabbard-128	128	-99	-102.1	1.68, 107.7	187.7 / 133.9	2.36, 368.4	432.4 / 248.2

Table 8: Minal-coded sets: exact DFR and the cheapest first failure under a query cap. S is the squared-norm tilt of the ciphertext secrets (s', e' for Rudraksh2; s' for Scabbard) and “precomp” its $\log_2$ cost.

Scabbard-128 ($\ell = 9, n = 64, \epsilon_q = 14, \epsilon_p = 10, \epsilon_t = 3, \eta = 2, \beta = 1030$). The exact DFR is $2^{-102.1}$, against the claimed 2^{-99} . The gap to an isotropic Gaussian of the same variance (about 2^{-67} in a second-moment screen) is the bounded support of the two rounding errors and of ρ . Without a query cap a failure costs 2^{102} , below the level 128 and below the Core-SVP value 127.6; this is the comparison the specification itself makes when it writes that $2^{64} \times 2^{-99}$ is negligible. Under a 2^{80} cap a tilt of s' to $S = 1.68$ costs 2^{108} offline and a classical total of 2^{188} (quantum 2^{134}); under a 2^{64} cap $S = 2.36$ costs 2^{368} offline. The encapsulation secret controls only about half of the noise variance, so the tilt is expensive.

The remaining Rudraksh2 sets claim DFRs $2^{-100}, 2^{-161}, 2^{-167}, 2^{-160}, 2^{-160}$ and the remaining Scabbard sets claim 2^{-125} ; none of them meets a 2^{80} conditional failure rate on a legal ciphertext at a cost below the level, and they are outside this paper.

Set	mechanism	level	claimed	ours	no cap	2^{80} cap	2^{64} cap	Core-SVP
DTRU-648	correlated blocks	128	-146.8	-85.4	85	84	86	164.7
DTRU-768	correlated blocks	192	-185.6	-97.9	98	86	87	199.4
DTRU-1024	correlated blocks	256	-190.5	-104.6	105	99	103	277.7
DTRU-Prime	correlated blocks	256	-180.7	-96.3	96	80	80	292.6
DTRU-1536	correlated blocks	384	-195.5	-100.7	101	95	99	416.7
DTRU-2048	correlated blocks	512	-204.1	-110.8	111	116	130	512.5
Cheetah256	omitted compression	256	-176	-51.8	52	52	52	325.9
Cheetah128	omitted compression	128	-129	-79.1	79	79	113	148.3
DTRU-Light	(none)	128	-143.9	-137.4 / -121.7	122-137	125	130	132.3
Rudraksh2-128-I	bounded noise, exact	128	-100	-103.9	104	112	168	197.7
Scabbard-128	bounded noise, exact	128	-99	-102.1	102	188	432	127.6

Table 9: $\log_2$ of the classical work for a first decryption failure. “ours” is the recomputed $\log_2 \delta$ (DTRU: sample mean; Light: sample mean / normal tail). The cap columns include the offline search. Core-SVP is the classical estimator value for key recovery. The first block is Class I (first-failure cost below the nominal level); the second block is Class II (on the line, or below the level only without a query cap).

Part IV

Summary and discussion

16 All affected parameter sets

Table 9 collects the first-failure costs. Eight parameter sets have a first-failure cost below their nominal level under a 2^{80} query cap, by two mechanisms; three more sit at the line.

17 Discussion

Three ways a DFR computation goes wrong. DTRU computes the per-coefficient variance correctly but treats the 16 coefficients of a decoding block as independent; in $x^n - x^{n/2} + 1$ the block is made of ring-paired coefficients and the repetition code places the same bit on both members of each pair. Cheetah models the right kind of tail but on the wrong noise: the decompressed public key carries a rounding error that multiplies r , and the v -compression error is a known shift, not a worst case. Rudraksh2 and Scabbard evaluate a two-dimensional FFT of the pair; we recover values a few bits below their claims. The remaining finding for those two sets is the query-cap arithmetic: a 2^{80} budget plus a modest offline tilt puts Rudraksh2-128-I below the classical 128-bit line, and Scabbard-128 without a cap is already below 128 by the specification’s own $1/\delta$ comparison.

Fixes. For DTRU the correct computation is cheap: form the 16×16 conditional covariance Σ_I per block and evaluate the half-space union of Lemma 1; at the design stage the coincidence can be removed by decoupling the repetition offset from the ring pairing. For Cheetah, $d_v = 4$ and $d_b = 10$ are the two parameters to revisit; the exact tail of Section 11 can replace the Gaussian, and $\langle e_b, r \rangle$ belongs in the noise. For the Minal schemes the exact pair tail is already close to the claim; what changes the level is whether $1/\delta$ or a 2^{80} total is the figure of merit.

Security levels. In the DFR metric, six of seven DTRU sets miss their nominal level by 43–400 bits, Cheetah256 by about 200 bits and Cheetah128 by 50 bits. Rudraksh2-128-I under a 2^{80} cap sits 16 bits below 128 classically and 16 bits above the 80-bit quantum target. Scabbard-128

without a cap is 26 bits below 128. The FO advantage term $(q_H + q_D)\delta$ becomes constant at $q_D \approx 2^{52}-2^{111}$ for the Class I sets. None of this yields a practical attack.

Limitations. Our DTRU estimate of $\mathbb{E}[p]$ is a tail quantity: the sample mean of 200 ciphertexts and the normal-tail extrapolation differ by 1–2 bits for DTRU-648 and -2048 and by 12–16 bits for DTRU-768, -Prime and -Light; narrowing this requires importance sampling of ciphertexts tilted towards large top eigenvalues. Failure probabilities are averaged over keys; weak keys were not studied. The layout of DoubleE₈ for DTRU-Prime is our assumption. For Cheetah the value at $q/4$ is an exact computation under the independence of coefficients that the designers also assume; the full-size check reaches $|E| \geq 800$ (2^{-11} to 2^{-14}) and the scaled-down check reaches the decoder. For the Minal schemes the pair tail is a radial approximation of a polygonal cell, checked on toys whose failure rates are 2^0 to 2^{-2} , so a few bits of geometric residual remain at the operating point; the squared-norm tilt is evaluated on a refined S search, and the reported costs are upper bounds within a few bits. The directional-boosting figures for DTRU are single-hyperplane conditional estimates, not a re-implementation of [7], and no lattice reduction on a failure-augmented instance was run for any scheme.

18 Conclusion

For lattice KEMs that decode blocks of coefficients with an error-correcting code, or that compress keys and ciphertexts, the DFR analysis must use the real decoder, the real noise and the exact tail. Correlations inside a decoding block raise the failure rate of six DTRU parameter sets by 50–110 bits; an omitted public-key compression term and a worst-case treatment of the ciphertext compression hide failure rates of 2^{-52} and 2^{-79} in Cheetah256 and Cheetah128; the exact pair tails of Rudraksh2-128-I and Scabbard-128 sit a few bits below the claims, and a 2^{80} query budget puts the former below the classical 128-bit line. The computations that expose this use 16×16 covariance matrices, tilted one-dimensional FFTs, the Voronoi radii of the schemes’ own decoders, and can be added to any DFR script.

Reproducibility. All scripts (`dfr_all_sets.py`, `ml_decoder_check.py`, `exact_cgf_check.py`, `recover.py` for DTRU; `exact_tails.py`, `cheetah_toy.py`, `boost_screen.py` for Cheetah, Rudraksh2 and Scabbard) and their JSON outputs are provided with this paper. They depend only on NumPy and SciPy.

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